SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –Hackathon**	**Assessment:**	**Assessment Tool 3– Hackathon**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and interpret datasets using data preprocessing, statistical analysis, visualization, and machine learning techniques for effective data-driven insights and decision-making.		
Student Name:	B Tharun Kumar	**Reg. No.:**	192424356
Section / Batch:	AI-DATA SCIENCE	**Date:**	29/08/26

Code of Conduct

I B Tharun Kumar – 192424356 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Tharun Kumar

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

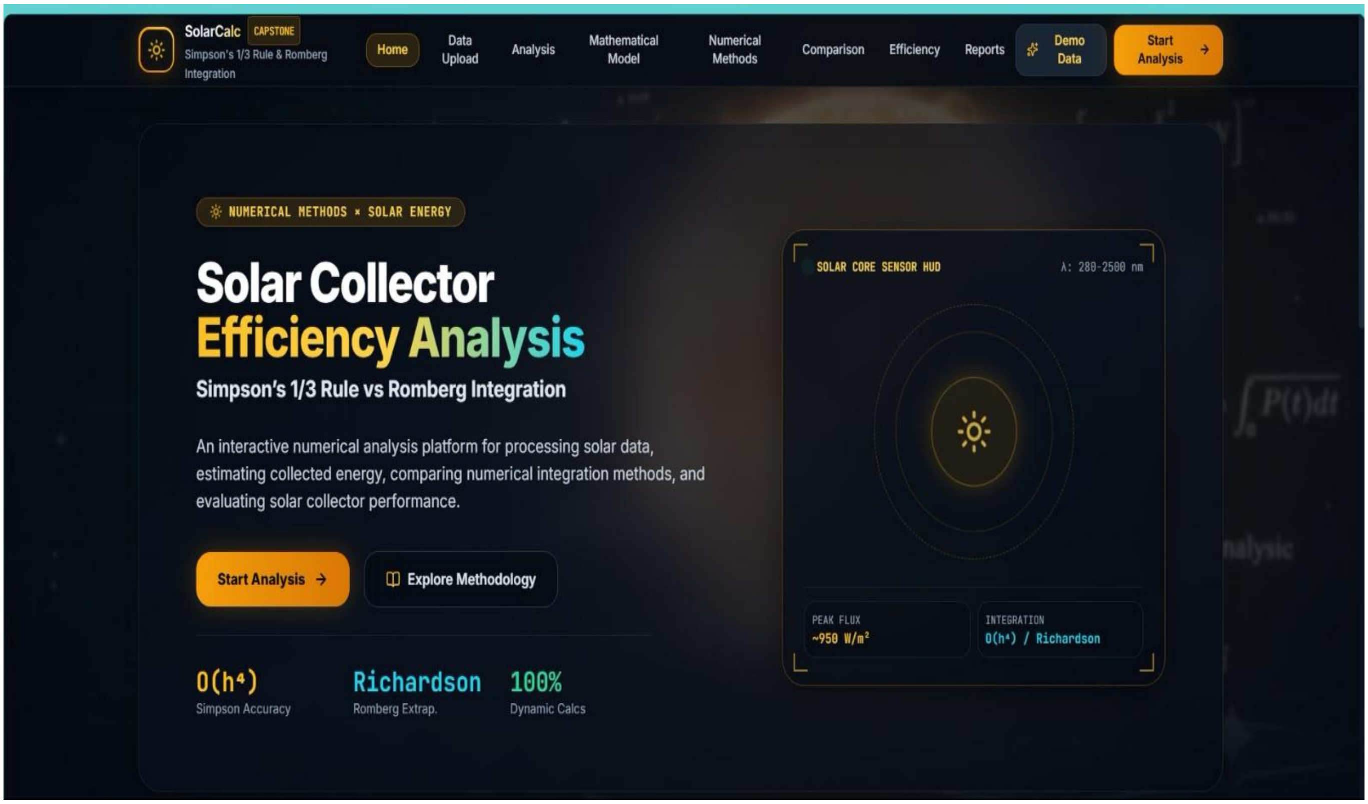
DSA0405 Fundamentals of Data Science
CO5-AT3 – Hackathon

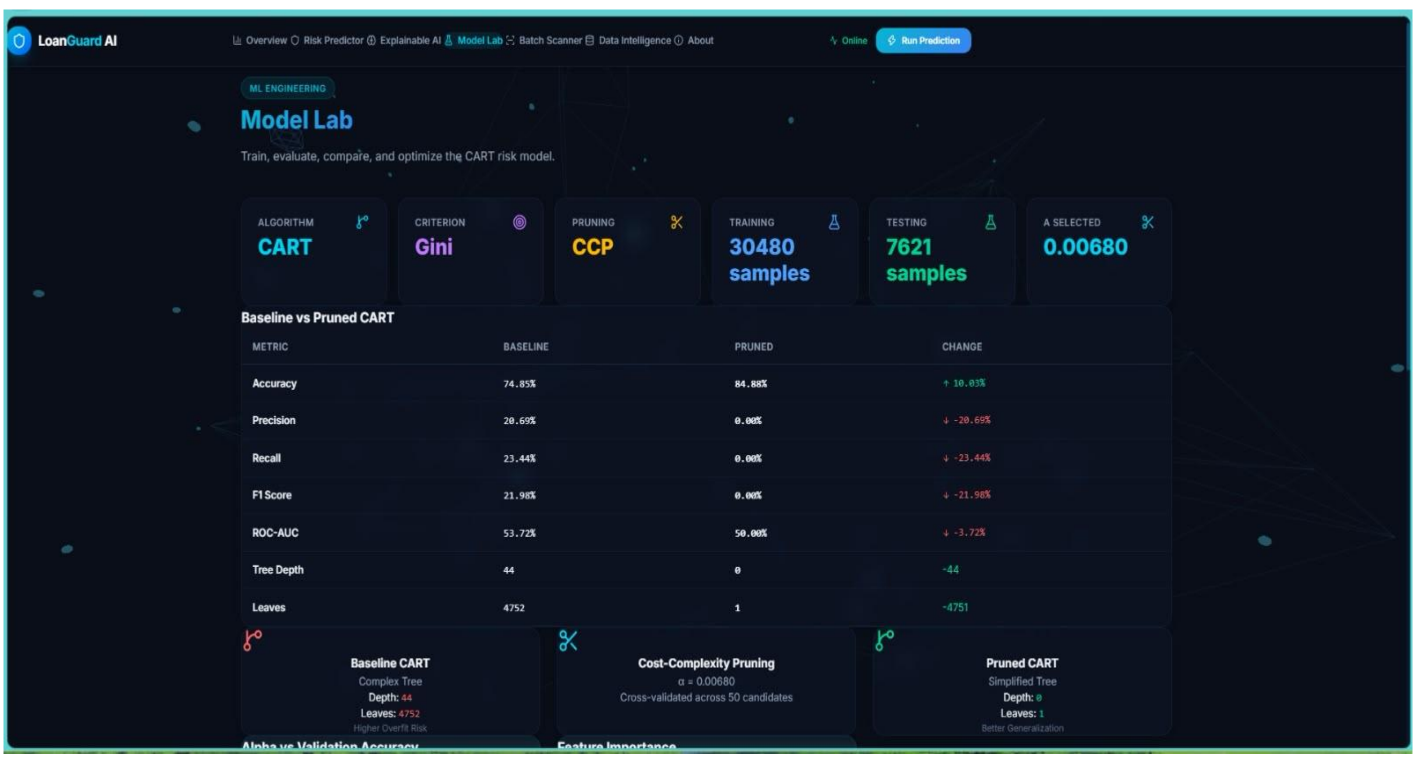
Code:

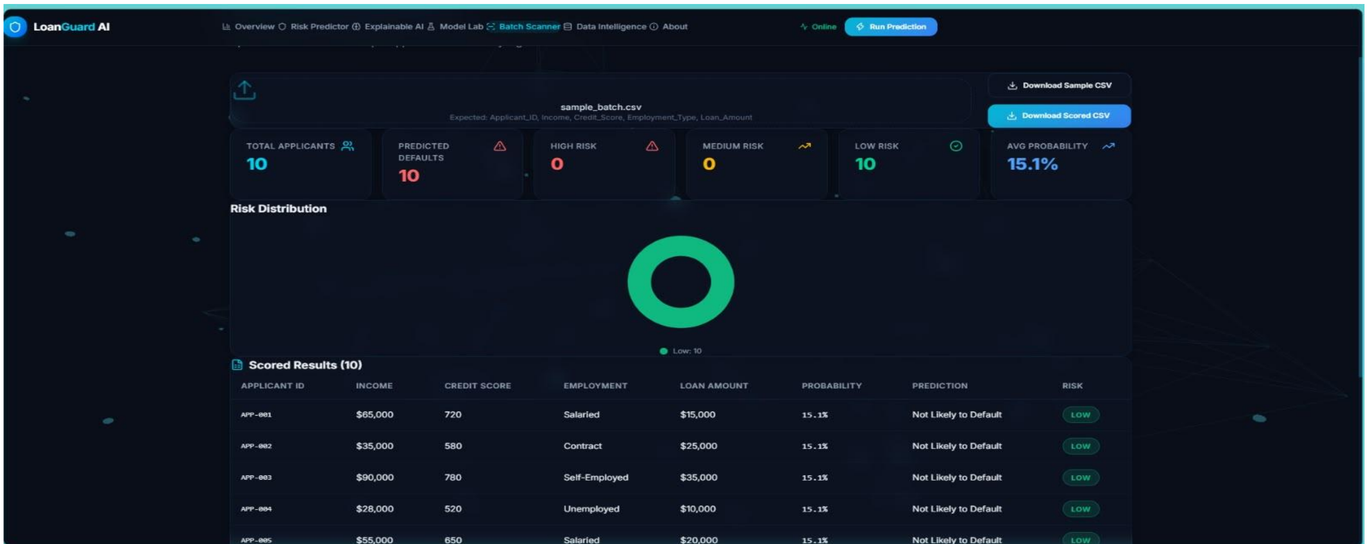
loanguard-ai > backend > app > services > ml_pipeline.py

```python
import os
import json
import logging
import numpy as np
import pandas as pd
import joblib
from datetime import datetime
from typing import Dict, Any, Tuple, List, Optional

from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, confusion_matrix, roc_curve,
    classification_report,
)
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

logger = logging.getLogger(__name__)

ARTIFACTS_DIR = os.path.join(os.path.dirname(os.path.dirname(os.path.dirname(__file__))), "artifacts")
DATA_DIR = os.path.join(os.path.dirname(os.path.dirname(os.path.dirname(__file__))), "data")


class MLPipeline:
    """Complete ML pipeline for loan default prediction using CART."""

    def __init__(self):
        self.baseline_model: Optional[DecisionTreeClassifier] = None
        self.pruned_model: Optional[DecisionTreeClassifier] = None
        self.preprocessor: Optional[ColumnTransformer] = None
        self.feature_names: List[str] = []
        self.original_feature_names: List[str] = []
        self.target_column: str = "Default"
```

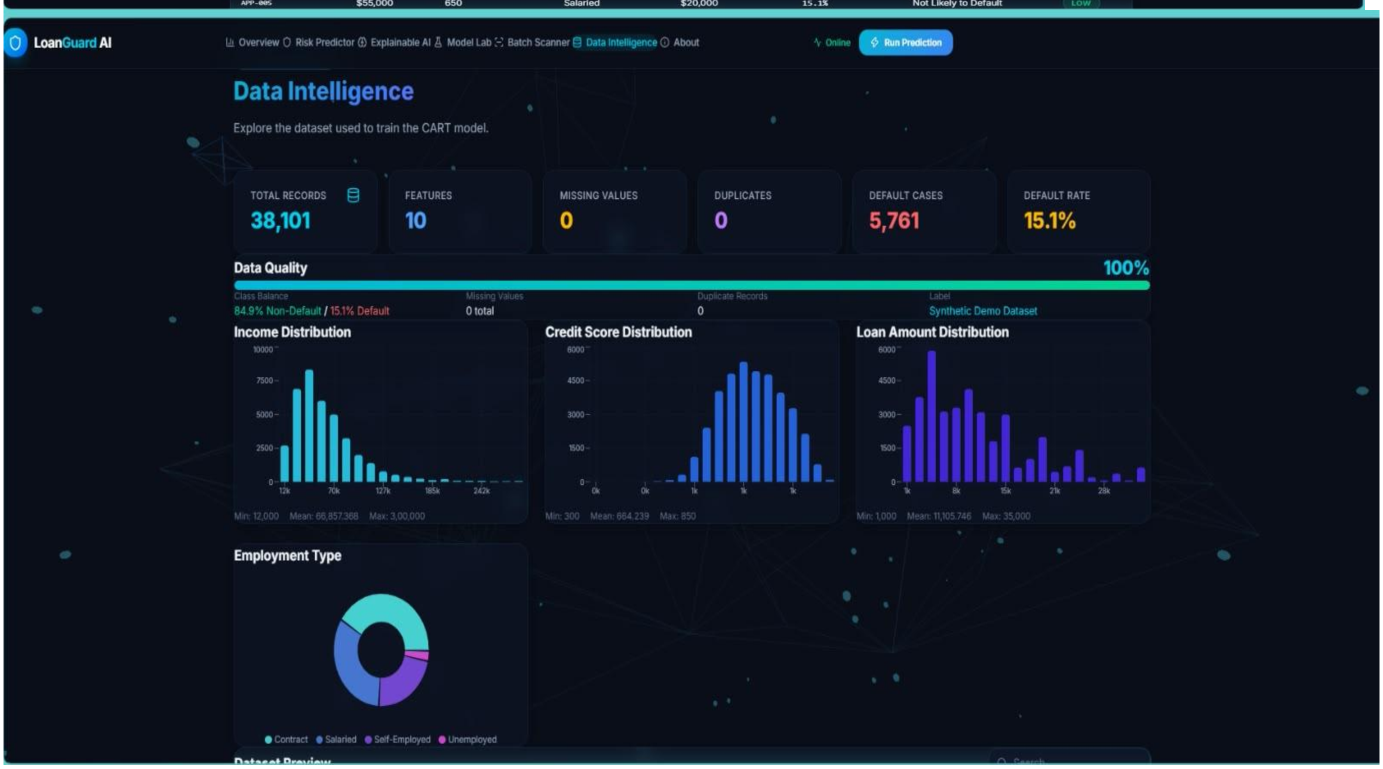
Implementation:







LoanGuard AI
Overview
Risk Predictor
Explainable AI
Model Lab
Batch Scanner
Data Intelligence
About
Online
Run Prediction
sample_batch.csv
Expected: Applicant_ID, Income, Credit_Score, Employment_Type, Loan_Amount
Download Sample CSV
Download Scored CSV
TOTAL APPLICANTS 10
PREDICTED DEFAULTS 10
HIGH RISK 0
MEDIUM RISK 0
LOW RISK 10
AVG PROBABILITY 15.1%
Risk Distribution
Low: 10
Scored Results (10)
APPLICANT ID | INCOME | CREDIT SCORE | EMPLOYMENT | LOAN AMOUNT | PROBABILITY | PREDICTION | RISK
APP-001 | $65,000 | 720 | Salaried | $15,000 | 15.1% | Not Likely to Default | LOW
APP-002 | $35,000 | 580 | Contract | $25,000 | 15.1% | Not Likely to Default | LOW
APP-003 | $90,000 | 780 | Self-Employed | $35,000 | 15.1% | Not Likely to Default | LOW
APP-004 | $28,000 | 520 | Unemployed | $10,000 | 15.1% | Not Likely to Default | LOW
APP-005 | $55,000 | 650 | Salaried | $20,000 | 15.1% | Not Likely to Default | LOW



LoanGuard AI
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About
Online
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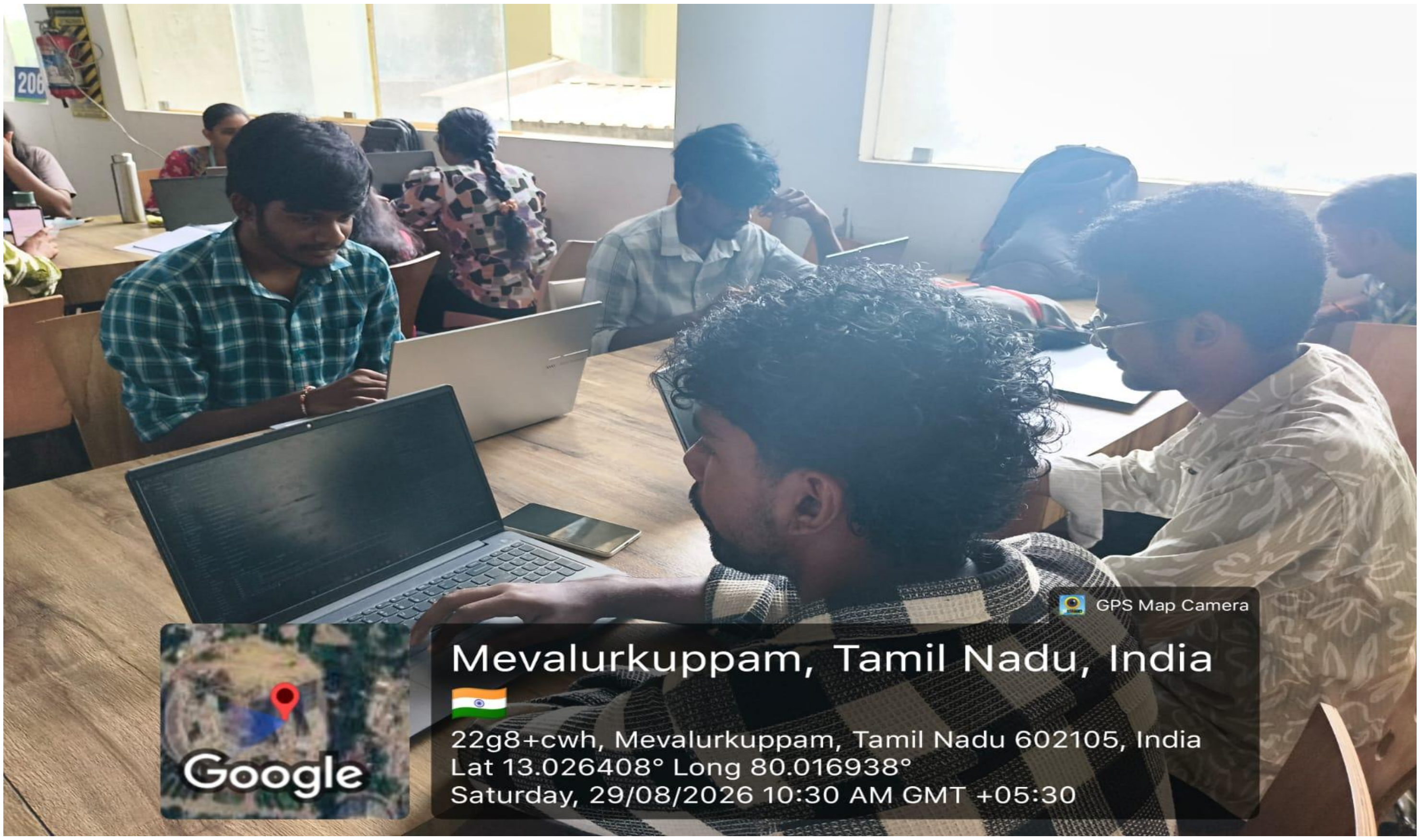
Data Intelligence
Explore the dataset used to train the CART model.
TOTAL RECORDS 38,101
FEATURES 10
MISSING VALUES 0
DUPLICATES 0
DEFAULT CASES 5,761
DEFAULT RATE 15.1%
Data Quality 100%
Class Balance
84.9% Non-Default / 15.1% Default
Missing Values
0 total
Duplicate Records
0
Label
Synthetic Demo Dataset
Income Distribution
Min: 12,000 Mean: 66,857.366 Max: 3,00,000
Credit Score Distribution
Min: 300 Mean: 664.239 Max: 850
Loan Amount Distribution
Min: 1,000 Mean: 11,105.746 Max: 35,000
Employment Type
Contract
Salaried
Self-Employed
Unemployed
Dataset Preview

Geotag: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